Weight and local gravity

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish mass from gravitational weight.

SUCCESS CRITERIA - I can...

1. Use mass in kg and force in N.
2. Calculate weight using the stated local g.
3. Explain how weight changes while mass stays the same.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “weight” and “mass” to compare your model or result. Sentence starters are optional.

weight: the gravitational force on an object

mass: a measure of how strongly an object resists acceleration

field: a description of an influence at each location

01 Warm-up: bring back an earlier idea

A simplified model predicts position (-50, 70) km with bounds ±5 km in x and ±7 km in y. Write a short briefing giving both intervals and two reasons it is not a real asteroid collision forecast.

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $F_g = mg$; $g = F_g/m$.
F_g is weight. Field strength in N/kg is numerically equal to free-fall acceleration in m/s².

Worked reasoning: A 3 kg object in a field of 4 N/kg has weight 3×4=12 N. Moving it to a 2 N/kg field halves weight to 6 N but leaves mass 3 kg.

Your turn: Draw weight arrows for the two places and label the unchanged quantity.

Sentence starter: The … in my model represents … . Its label shows … .

Location	Mass (kg)	g (N/kg)	Weight (N)
A	3	4	12
B	3	2	6

Marzano check - circle: ① ② ③

03 Read and interpret the model

The same 2 kg object visits worlds with g=10 and g=5 N/kg.
Compare its mass and weight.

Sentence starter: Both … . They differ in … because … .

World	g (N/kg)
A	10
B	5

Marzano check - circle: ① ② ③

04 Apply the physics

An astronaut has mass 60 kg. Complete the location table using W = mg.
Explain which property of the astronaut stays the same.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Location	g (N/kg)	Weight (N)
Earth	9.8	____
Moon	1.6	____
Planet	____	240

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A scale reports 0 N for an astronaut in an orbiting spacecraft. Does that prove Earth exerts no gravitational force?
Explain.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A 4 kg cart has 6 N right and 38 N left.
Find acceleration. What else must you know to decide if it speeds up?
Draw the forces on the chosen object and label their sources.

Word help: acceleration = rate of change of velocity

Sentence starter: I found … [unit]. This value means … in this situation.

Add and label force arrows.

Marzano check - circle: 1 2 3

07 Exit check

An object weighs 49 N where g=9.8 N/kg.
Find its mass and weight where g=2 N/kg.

Sentence starter: I found … [unit]. This value means … in this situation.

Add and label force arrows.

Draw and label the gravitational force.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

An object weighs 72 N at one location and 18 N at another where g=3 N/kg.
Find its mass and the first field strength.

Sentence starter: I found … [unit]. This value means … in this situation.

Case	Mass (kg)	g (N/kg)	Weight (N)
1	____	____	____
2	____	____	____

Marzano check - circle: 1 2 3